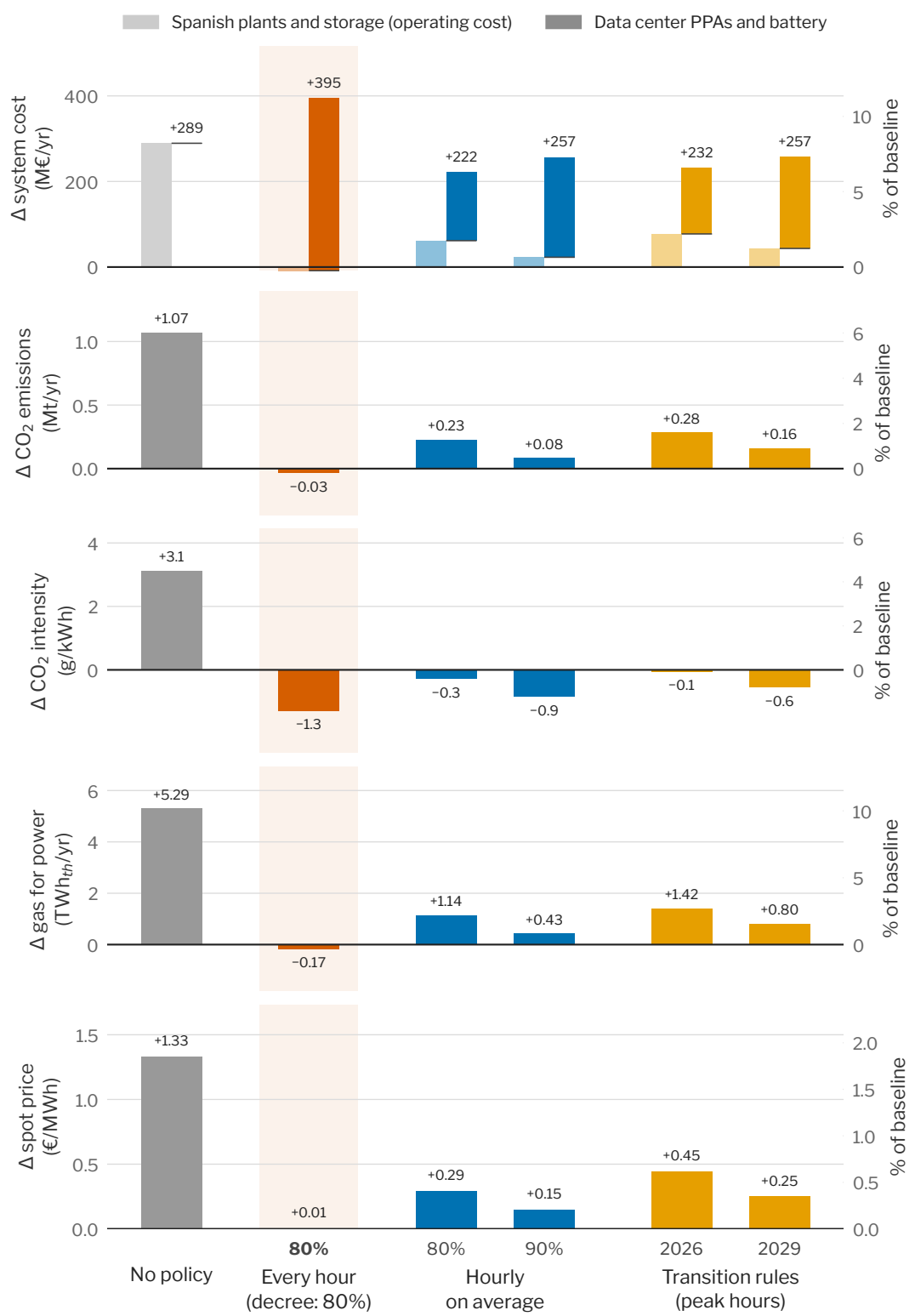

Change in the Spanish power system with a 500 MW data center in Aragon against a baseline without it, absolute (left axes) and in percent of the baseline level (right axes). System cost: operating cost of Spanish plants and storage (light), then the data center's PPA and battery costs on top (dark), ending at the total. CO2: fuel burnt in Spanish plants times its carbon content. CO2 intensity: Spanish CO2 per kWh of Spanish generation, the data center's PPAs included. Gas: fuel burnt in Spanish gas-fired power plants (CCGT, OCGT). Spot price: hourly nodal prices weighted by Spanish demand, the data center excluded. PPAs anywhere in mainland Spain. Transition rules: the revised draft's phase-in for data centers in the automatic load-reduction scheme, a floor only in weekday tariff peak hours: 60% in P1 hours under the 2026 rules (to 2028), then 70% in P1 and 60% in P2 hours under the 2029 rules; all other hours are unconstrained (P1: 9–14 h and 18–22 h on weekdays in January, February, July and December; P2: 8–9, 14–18 and 22–24 h on those days, and 9–14 h and 18–22 h on weekdays in March and November; local time, national holidays excluded).